

The least stable threshold for four-value generalized Schur numbers

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Abstract

Let $S_2(k; 3)$ be the least N such that every two-coloring of $\{1, \dots, N\}$ admits a monochromatic solution of $x_1 + \dots + x_k = x_{k+1}$ with exactly four distinct positive values; repetitions among the summands are allowed. We prove $S_2(k; 3) = k^2 + 4k + 3$ for every integer $k \geq 3$. Thus the least admissible stable threshold is 3. The proof makes Gaiser's color-forcing argument explicit for $k \geq 10$ and completes the seven remaining cases with exact Boolean unsatisfiability certificates. The certificates and two standalone verification implementations accompany the paper.

1 Statement and relation to prior work

Write $[N] = \{1, \dots, N\}$. For $k \geq 3$, a *four-value solution* is a positive-integer solution of

$$x_1 + \dots + x_k = x_{k+1} \tag{1}$$

in which the set $\{x_1, \dots, x_{k+1}\}$ has cardinality exactly four. The summands need not be distinct. A two-coloring is any map $[N] \rightarrow \{R, B\}$, without a surjectivity requirement. It is *avoiding* if it contains no monochromatic four-value solution. We use Gaiser's notation $S_2(k; 3)$ for the smallest N with no avoiding two-coloring.

Theorem 1. *For every integer $k \geq 3$,*

$$S_2(k; 3) = k^2 + 4k + 3.$$

Consequently, the least integer $K \geq 3$ such that this equality holds for every $k \geq K$ is $K = 3$.

Gaiser [1, Theorem 1.1] proves the lower bound in this formula for every $k \geq 3$ and equality for all sufficiently large k . His Open Question 6.4 asks for the least stable threshold for a fixed number of distinct values. Theorem 1 determines that threshold in the four-value case. The lower coloring and the general color-forcing method are due to Gaiser [1, Sections 3 and 5]; we reproduce their specialization to give the full argument with the explicit sufficient cutoff 10. The finite certificates then establish equality throughout the remaining admissible range. No conclusion for other numbers of distinct values is asserted here.

The use of Boolean encodings with independently replayable refutations also occurs in Zeng's work on the Erdős–Székere problem [2], which provides adjacent computational methodology. It is not a dependency for the Schur-number result. The prior-work comparison is based on Gaiser's version 1 and bounded primary-source searches on 7 September 2026; no equivalent determination of this threshold was found in those searches. This is not an exhaustive priority claim.

2 Supports and the lower bound

Because all summands in (1) are positive, their sum is strictly greater than each summand. Thus a four-value solution is equivalent to integers

$$1 \leq a < b < c < d \leq N, \quad p, q, r \geq 1, \quad p + q + r = k, \quad pa + qb + rc = d. \quad (2)$$

Indeed, the three summand values are a, b, c , with their positive multiplicities p, q, r ; conversely these data give the required solution. In particular, its four-element support, rather than a choice of multiplicities representing that support, determines the coloring constraint.

Throughout the proof put

$$F = k^2 + 4k + 3, \quad Q = k^2 + 3k + 3.$$

Lemma 2. *For every integer $k \geq 3$ there is an avoiding two-coloring of $[F - 1]$.*

Proof. Color $[1, k + 2] \cup [Q, F - 1]$ red and $[k + 3, Q - 1]$ blue. If three different red summand values all lie in $[1, k + 2]$, a sum with positive multiplicities totaling k is at least $(k - 2) \cdot 1 + 2 + 3 = k + 3$ and at most $k(k + 2) < Q$; it is therefore blue. If a red summand is at least Q , the presence of three different summand values makes the sum at least $Q + (k - 2) \cdot 1 + 2 = F$, outside $[F - 1]$. Finally, three different blue values give a sum at least

$$(k - 2)(k + 3) + (k + 4) + (k + 5) = Q,$$

which is not blue. These alternatives exclude every monochromatic support of the form (2). $\square$

3 The upper bound for all $k \geq 10$

Proposition 3. *For every integer $k \geq 10$, every two-coloring of $[F]$ contains a monochromatic four-value solution.*

Proof. Suppose instead that the coloring is avoiding. We use the following forcing rule: if three distinct values have one color, then any of their sums with positive multiplicities totaling k must have the other color, provided the sum lies in $[F]$.

Some color occurs at least three times in $[5]$. Select the lexicographically least increasing monochromatic triple $u = (u_1, u_2, u_3)$ and call its color red. Every integer less than u_3 omitted from u is blue: an omitted red integer would give an earlier red triple. There are ten possibilities for u . We treat four separately and the other six together. Positivity, distinctness and the common range of all identities are verified at the end.

The triple $(1, 2, 3)$. For $0 \leq j \leq 4$, the identities

$$(k - 2 - j) \cdot 1 + (1 + j) \cdot 2 + 3 = k + 3 + j$$

force $k + 3, \dots, k + 7$ blue. Hence

$$\begin{aligned} (k - 2)(k + 3) + (k + 4) + (k + 5) &= Q, \\ (k + 3) + (k - 3)(k + 4) + 2(k + 6) &= F \end{aligned}$$

force Q and F red. But $(k - 2) \cdot 1 + 2 + Q = F$ is then a red four-value solution.

The triple $(1, 3, 5)$. The values 2 and 4 are blue. The identities

$$\begin{aligned} (k - 2) \cdot 1 + 3 + 5 &= k + 6, \\ (k - 2) \cdot 2 + 4 + (k + 6) &= 3k + 6, \\ (k - 2) \cdot 1 + 3 + (3k + 6) &= 4k + 7, \\ (k - 2) \cdot 2 + 4 + (4k + 7) &= 6k + 7 \end{aligned}$$

force their outputs blue, red, blue and red, respectively. The final identity $(k-3) \cdot 3 + 2 \cdot 5 + (3k+6) = 6k+7$ is a red four-value solution.

Six triples with an adjacent pair. These are

$$(1, 2, 4), (1, 2, 5), (2, 3, 4), (2, 3, 5), (3, 4, 5), (2, 4, 5).$$

Let v be the least positive integer missing from u and let j be the first index with $u_{j+1} = u_j + 1$. Then $v \leq 3$, $v < u_3$, and v is blue. Put

$$\begin{aligned} U &= u_1 + u_2 + u_3, & A &= U + (k-3)u_j, \\ V &= (k-2)v + 2A + 1, & W &= (k-3)v + 3A + 1. \end{aligned} \tag{3}$$

One copy of each red u_i and $k-3$ extra copies of u_j force A blue. Replacing one of the extra copies by u_{j+1} forces $A+1$ blue. Consequently

$$(k-2)v + A + (A+1) = V, \quad (k-3)v + 2A + (A+1) = W$$

force V, W red.

In the first five triples $j = 1$. Set $h = u_3 - v$. Then $1 \leq h \leq 4$, and

$$(k-2-h)u_1 + (1+h)u_2 + V = W \tag{4}$$

is a red solution. To verify the identity, use $u_2 = u_1 + 1$ to write its left-hand side minus V as $(k-1)u_1 + 1 + h = U + (k-3)u_1 - v = A - v = W - V$. For $(2, 4, 5)$ we have $v = 1$, $j = 2$, $A = 4k-1$, $V = 9k-3$, $W = 13k-5$, and the red solution is instead

$$(k-3) \cdot 4 + 2 \cdot 5 + V = W.$$

The triple $(1, 4, 5)$. Here 2, 3 are blue. The identities

$$(k-2) \cdot 1 + 4 + 5 = k+7, \quad 2 + (k-2) \cdot 3 + (k+7) = 4k+3$$

force $k+7$ blue and $4k+3$ red. But $1 + (k-7) \cdot 4 + 6 \cdot 5 = 4k+3$ is a red solution.

The triple $(1, 3, 4)$. The value 2 is blue. The four identities

$$\begin{aligned} (k-2) \cdot 1 + 3 + 4 &= k+5, & (k-3) \cdot 1 + 2 \cdot 3 + 4 &= k+7, \\ (k-4) \cdot 1 + 3 \cdot 3 + 4 &= k+9, & (k-4) \cdot 1 + 2 \cdot 3 + 2 \cdot 4 &= k+10 \end{aligned}$$

force blue outputs. Next

$$\begin{aligned} (k-2) \cdot 2 + (k+5) + (k+7) &= 4k+8, \\ (k-2) \cdot 2 + (k+9) + (k+10) &= 4k+15 \end{aligned}$$

force red outputs, and

$$\begin{aligned} (k-2) \cdot 1 + 3 + (4k+8) &= 5k+9, \\ 3 + (k-2) \cdot 4 + (4k+15) &= 8k+10 \end{aligned}$$

force blue outputs. Finally $(k-2) \cdot 2 + (k+5) + (5k+9) = 8k+10$ is a blue solution.

Validity of all forcing steps. Every displayed multiplicity is a positive integer and each group totals k . The smallest possibly occurring count is $k-7$, which is positive for $k \geq 10$. In (4) the first count is at least $k-6$. The small input values are explicitly different. In the six adjacent-pair cases, $v < A < A+1 < V < W$ and $V > u_3$, so both the intermediate and

final triples have different values. In the remaining cases, the displayed larger inputs exceed all small inputs, and any pair of larger inputs used together has nonzero difference. Since the sum includes each of the three distinct inputs with positive multiplicity, it is strictly greater than each of them.

The first case uses values at most F ; in particular $k + 7 < Q < F$. The second uses values at most $6k + 7$. In the adjacent-pair cases, $U \leq 12$, $v \leq 3$ and $u_j \leq 4$, so

$$W = (k - 3)v + 3U + 3(k - 3)u_j + 1 \leq 15k - 8.$$

The last two cases use values at most $4k + 3$ and $8k + 10$. Writing $k = 10 + t$, $t \geq 0$, gives

$$\begin{aligned} F - (6k + 7) &= t^2 + 18t + 76, \\ F - (15k - 8) &= t^2 + 9t + 1, \\ F - (8k + 10) &= t^2 + 16t + 53. \end{aligned}$$

All three expressions are positive, and $4k + 3 \leq 8k + 10$. Thus every forcing step takes place in $[F]$ and concerns exactly four distinct values. All ten initial triples lead to a contradiction. $\square$

4 Exact certification for $3 \leq k \leq 9$

For fixed k, N , form a four-uniform hypergraph with vertex set $[N]$ whose edges are precisely the supports in (2). Give vertex i a Boolean variable X_i , with true denoting red. For each edge $\{a, b, c, d\}$ include the two clauses

$$(X_a \vee X_b \vee X_c \vee X_d), \quad (\neg X_a \vee \neg X_b \vee \neg X_c \vee \neg X_d). \quad (5)$$

Their conjunction is satisfied exactly by avoiding colorings. We also add the unit clause X_1 . This preserves satisfiability, because complementing every variable preserves all clause pairs, so any avoiding coloring can be exchanged with one in which 1 is red. There is no additional symmetry restriction.

Lemma 4 (Finite certificate computation). *For each $k \in \{3, 4, 5, 6, 7, 8, 9\}$, the formula just defined with $N = F$ is unsatisfiable.*

Certificate and verification. The accompanying directory `certificates/` contains the formulas `k3.cnf` through `k9.cnf`, with corresponding `.lrat` refutations. Their sizes and checked deduction counts are listed in Table 1. We describe both the exact encoding check and the inference rule used to verify these finite certificates.

To generate the edge set, enumerate $1 \leq a < b < c < N$ and positive p, q, r with $p + q + r = k$, retaining $d = pa + qb + rc \leq N$. The only pruning is $(k - 2)a + b + c > N$, since this is the least sum using the three values. Deduplicating supports preserves all the constraints in (5).

Two separately implemented reconstructions check completeness against the saved clauses. The first starts at $T_0 = \{a + b + c\}$, discarding values above N , and iterates

$$T_{j+1} = \{s + x : s \in T_j, x \in \{a, b, c\}, s + x \leq N\}.$$

The set T_{k-3} consists exactly of the sums with positive multiplicities totaling k . Truncation loses no possible final sum because all added terms are positive. The second loops over $a < b < c$, $d \geq ka + (b - a) + (c - a)$, and $1 \leq r \leq k - 2$, solving

$$q(b - a) = d - ka - r(c - a).$$

Table 1: Exact finite certificates. Clause counts include the unit X_1 .

k	F	Distinct supports	CNF clauses	RUP additions
3	24	274	549	68
4	35	1,317	2,635	55
5	48	4,046	8,093	52
6	63	9,941	19,883	53
7	80	21,325	42,651	46
8	99	41,295	82,591	62
9	120	74,388	148,777	47

An integral q satisfying $1 \leq q \leq k - r - 1$ gives, and is given by, the positive multiplicities $(p, q, r) = (k - q - r, q, r)$. Both reconstructions match all saved positive and negative clauses and the single symmetry unit exactly.

Each added clause C in the refutation has a list of positive identifiers of active clauses. Assume every literal of C is false. In the listed order, each referenced clause must be unit under the current assignment, forcing its remaining literal, until a referenced clause is false. This final conflict proves that the active clauses entail C . This rule is reverse unit propagation (RUP). The verifiers reject unknown or deleted references, satisfied or nonunit hints, missing conflicts and negative hints. No more general RAT inference is needed by these certificates. Clause deletions do not affect soundness: by induction every remaining active clause is a consequence of the original formula. A verified final empty clause therefore proves unsatisfiability.

Both implementations verify all seven empty clauses. The second replay checks 383 clause additions and 1,641 propagation hints in total. Its controls reject eight invalid deductions and accept two valid ones. These exact deductions establish the asserted finite unsatisfiability statements. $\square$

Proof of Theorem 1. Lemma 2 gives $S_2(k; 3) \geq F$ for every $k \geq 3$. Lemma 4 gives the reverse bound for $3 \leq k \leq 9$; Proposition 3 gives it for $k \geq 10$. These ranges cover every admissible k , with no monotonicity assumption in k . Hence equality holds throughout, and 3 is the least admissible stable threshold. $\square$

5 Reproducibility and assistance

The accompanying `supplement/` directory contains the fourteen CNF/LRAT files, byte-for-byte copies of the two verification implementations, an exact symbolic record of the forcing identities, and a runner. From that directory, execute

```
python3 run_verification.py
```

Only Python’s standard library is required. The runner reconstructs the finite edge sets by both methods, replays both certificate checkers, and checks the 55 identities across the ten branches using integer polynomial arithmetic in $t = k - 10$. It also checks all 32 colorings of [5] against the initial-triple premises. The written argument proves the infinite range; no numerical sampling is used to infer an assertion for all k .

Kissat 4.0.4 produced the original DRAT refutations, and `drat-trim` checked them and converted them to LRAT. Neither program is required to verify the saved certificates. Checksums, exact commands and original tool provenance are supplied with the computational

materials. The verification implementations were separately written, including a fresh model-context replay; neither implementation is certified in a proof assistant.

OpenAI Codex assisted with the source comparison, specialization of the forcing argument, implementation and execution of the exact computations, verification, and manuscript preparation. The separate model-context check is computational assistance, not human peer review. The finite part of the proof depends on the supplied exact certificates and their replay; it is not a claim of proof-assistant formalization. The mathematical scope is precisely the two-color, four-value statement in Theorem 1.

References

- [1] Collier Gaiser, *Restricted generalized Schur numbers*, arXiv:2608.08789v1 (2026). <https://arxiv.org/abs/2608.08789v1>.
- [2] Zijian Zeng, *One-Sided Parity Compression for Certified SAT Attacks on the 33-Point Erdős–Szekeres Problem*, SSRN working paper 7334758 (2026). <https://doi.org/10.2139/ssrn.7334758>.